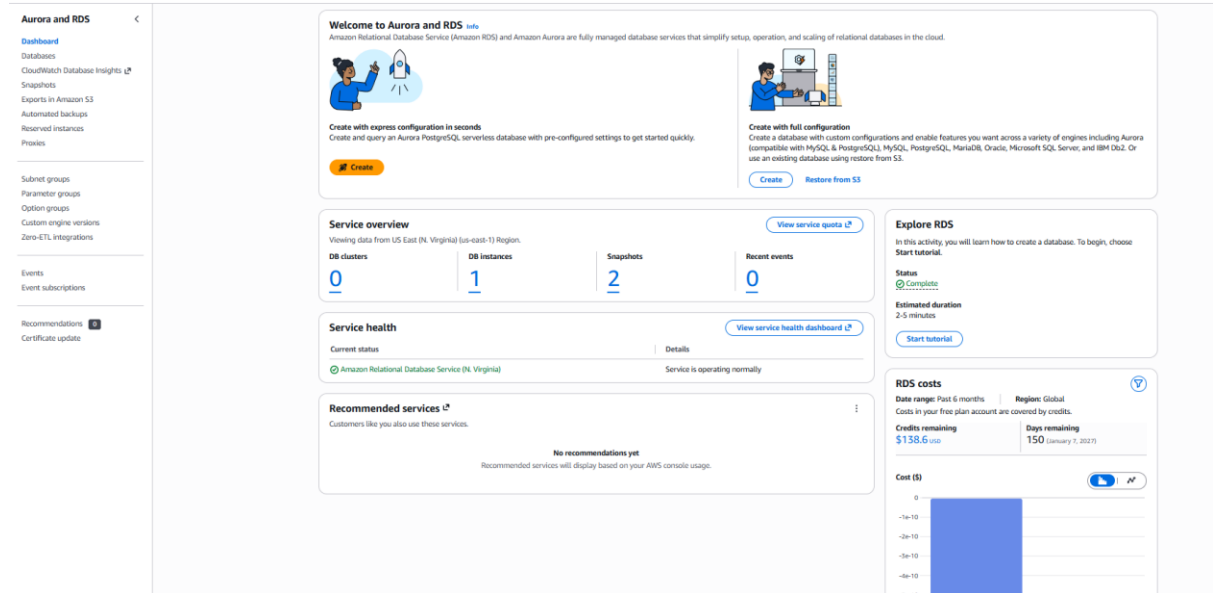


Cloud Databases (Amazon RDS) Labs

Lab 1 - Exploring Amazon RDS

- AWS RDS Dashboard



Lab 2

- Amazon RDS support

Lab 3 - Create Your First Amazon RDB Database

Method

- **Standard Create (Full configurations – no standard creates)**

Engine

- **MySQL** | Template: **Free Tier**

Database Identifier

- **student-db**

Master Username | Password

- **admin** | **1kFGvjx8o6pC**

Database Name:

- **academy**

Public Access

- **No**

Create database [Info](#)

Free plan has access to limited features and resources

The free plan limits the features and resources that are available for RDS and Aurora databases. Upgrade your account plan to remove all limitations. [Learn more](#)

[Upgrade plan](#)

Engine options

Engine type [Info](#)

☐ Aurora (MySQL Compatible)



☐ Aurora (PostgreSQL Compatible)



☒ MySQL



☐ PostgreSQL



☐ MariaDB



☐ Oracle

ORACLE

☐ Microsoft SQL Server



☐ IBM Db2

IBM Db2

Choose a database creation method

☒ Full configuration

You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ Easy create

Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Templates

Choose a sample template to meet your use case.

☐ Production

Use defaults for high availability and fast, consistent performance.

☐ Dev/Test

This instance is intended for development use outside of a production environment.

☒ Free tier

Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS.

DB instance identifier [Info](#)

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

student-db

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Credentials Settings

Master username [Info](#)

Type a login ID for the master user of your DB instance.

admin

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management

You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - most secure

RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed

Create your own password or have RDS create a password that you manage.

☐ Auto generate password

Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

Password strength **Very strong**

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / * @

Confirm master password [Info](#)

Public Access

- **No**

Public access [Info](#)

☐ Yes

RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☒ No

RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

Database created

Databases (1)

Filter by databases

[Group resources](#) [Modify](#) [Actions](#) [Create database](#)

DB identifier	Status	Role	Engine	Upgrade rollout ...	Region ...	Size	Recommendations	CPU	Current...	Mainte...	VPC
student-db	Backlog	Instance	MySQL Community	SECOND	us-east-1a	db.t4g.micro		5.44%	0 Conr	none	vpc-0eb534171

Lab 4 – Explore the Database Configuration

Audit database instance metadata, networking topology & storage settings

Engine

- **MySQL Community**

Availability Zone

- **us-east-1a**

student-db

Modify

Actions

Summary

DB identifier

student-db

CPU

5.52%

Status

Available

Class

db.t4g.micro

Role

Instance

Current activity

0 Connections

Engine

MySQL Community

Region & AZ

us-east-1a

Recommendations

Storage

- **20 GiB**

VPC

- **vpc-0e9c534171f690a08**

Instance		Instance class		Primary storage		Monitoring	
Configuration							
DB instance ID student-db		Instance class db.t4g.micro		Encryption Enabled		Monitoring type Database Insights - Standard	
Engine version 8.4.9		vCPU 2		AWS KMS key aws/rds us		Database counter metrics Disabled	
RDS Extended Support Disabled		RAM 1 GiB		Storage type General Purpose SSD (gp2)		Enhanced Monitoring Disabled	
DB name -		Availability		Storage 20 GiB			
License model General Public License		Master username admin		Provisioned IOPS -			
Option groups default:mysql-8-4 		Master password *****		Storage throughput -			
Amazon Resource Name (ARN) arn:aws:rds:us-east-1:208073622332:db:student-db		IAM DB authentication Not enabled		Storage autoscaling Enabled			
Resource ID db-PN4AOC7NKA7UYVQZM4F3L3TQ6AA		Multi-AZ No		Maximum storage threshold 1000 GiB			
Created time August 04, 2026, 20:49 (UTC+02:00)		Secondary Zone -		Storage file system configuration Current			
DB instance parameter group default:mysql8.4 							
Deletion protection Disabled							
Architecture settings Non-multitenant architecture							

Endpoint

- **student-db.cyf4o0qgmrxy.us-east-1.rds.amazonaws.com:3306**

Connectivity & security		
Endpoint & port		
Endpoint student-db.cyf4o0qgmrxy.us-east-1.rds.amazonaws.com		
Port 3306		
Networking		
Availability Zone us-east-1a		
VPC vpc-0e9c534171f690a08		
Subnet group default-vpc-0e9c534171f690a08		
Subnets subnet-08d027355e39ffb61 subnet-00836247e66e3ae8b subnet-0f13cb28f16f1f0b subnet-0a600f4687a8d4977 subnet-08d3d81154d775f9d subnet-096e6f1df863e729c		
Network type IPv4		
Security		
VPC security groups default (sg-010a6cbff151c7547) 		
Publicly accessible No		
Certificate authority info rds-ca-rsa2048-g1		
Certificate authority date May 26, 2061, 01:34 (UTC+02:00)		
DB instance certificate expiration date August 04, 2027, 20:48 (UTC+02:00)		

Lab 5 – Understanding Backups

Are backups enabled?

- **Yes**

How many days are backups retained?

- **1 Day**

Why are backups important?

- *They guarantee business continuity, regulatory compliance & recovery against accidental data corruption or malicious deletion.*

student-db ⓘ Ⓐ Modify Actions

Summary

DB identifier student-db	Status Available	Role Instance	Engine MySQL Community	Recommendations
CPU 4.18%	Class db.mg.micro	Current activity 0 Connections	Region & AZ us-east-1a	

Connectivity & security | Monitoring | Logs & events | Configuration | Zero-ETL integrations | **Maintenance & backups** | Data migrations | Tags | Recommendations

Maintenance

Auto minor version upgrade Enabled	Maintenance window August 07, 2026 08:58 - 09:28 (UTC+02:00)	Pending maintenance none	Pending modifications
Upgrade rollout order second			

Pending maintenance (0) ⓘ Apply now Apply at next maintenance window

Description	Type	Status	Apply date
No pending maintenance available			

Backup

Automated backups Enabled (1 Day)	Latest restore time August 04, 2026, 21:03 (UTC+02:00)	Replicate to Region -
Copy tags to snapshots Enabled	Backup window 04:54-05:24 UTC (GMT)	Replicated automated backup -
Backup target AWS Cloud (US East (N. Virginia))		

Delete student-db instance

Permanently delete **student-db** DB instance. You can't undo this action.


Proceeding with this action will delete the instance with all its content and can affect related resources. [Learn more](#)

☒ Create final snapshot
Determines whether a final DB Snapshot is created before the DB instance is deleted.

Final snapshot name
The identifier of the new DB snapshot that is created.

student-db-snapshot

☒ Retain automated backups
Determines whether retaining automated backups for 1 day after deletion


You will be billed for retained backup storage at the rate described as 'Additional backup storage' found in [Backup Storage](#).

To avoid accidental deletion provide additional written consent.

To confirm deletion, type *delete me* into the field.

Cancel

Delete

Lab 6 – Create a Manual Snapshot

Actions → Take Snapshot → Name: student-db-snapshot

Snapshot

- ***Student-db-snapshot***

Take DB Snapshot

Preferences

To take a DB Snapshot, choose a database and name your DB Snapshot.

DB instance

DB instance identifier. This is the unique key that identifies a DB instance.

student-db

Snapshot name

Identifier for the DB Snapshot.

student-db-snapshot

Snapshot identifier is case insensitive, but stored as all lower-case, as in "mysnapshot". Cannot be null, empty, or blank. Must contain from 1 to 255 alphanumeric characters or hyphens. First character must be a letter. Cannot end with a hyphen or contain two consecutive hyphens.

Cancel

Take snapshot

Snapshot created

Successfully created snapshot student-db-snapshot. View details

Snapshots

Manual

System

Shared with me

Public

Backup service

Exports in Amazon S3

Manual snapshots (1)

Filter by manual snapshots

Snapshot name

Engine version

DB instance or cluster

Snapshot creation time

DB instance created time

Status

Progress

VPC

Snapshot type

Storage

☐	student-db-snapshot	8.4.9	student-db	August 04, 2026, 21:17 (UTC+02:00)	August 04, 2026, 20:49 (UTC+02:00)	Available...	Completed	vpc-0e9c534171f690a08	DB instance	20 GiB
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Snapshot details

student-db-snapshot

Details

ARN
arn:aws:rds:us-east-1:208073622332:snapshot:student-db-snapshot

Instance/Cluster Name
student-db

Master username
admin

DB snapshot name
student-db-snapshot

Snapshot type
manual

DB engine
mysql

DB engine version
8.4.9

License model
general-public-license

Architecture settings
Non-multitenant architecture

Option group
default:mysql-8-4

Zone
us-east-1a

KMS key ID
a3c1ed9c-496e-452e-9def-a808b3773d70

Source region
N/A

Snapshot Creation Time
August 04, 2026, 21:17 (UTC+02:00)

Original Snapshot Creation Time
August 04, 2026, 21:17 (UTC+02:00)

Instance/Cluster Creation
August 04, 2026, 20:49 (UTC+02:00)

Backup retention period
1 day

Backup window
04:54-05:24

VPC
vpc-0e9c534171f690a08

Status
Available

Storage type
General Purpose SSD (gp2)

DB storage
20 GiB

IOPS
-

Storage throughput
0

Port
3306

Time zone

Dedicated Log Volume
Turned off

Lab 7 Restore from a Snapshot

Select Snapshot → Choose: Restore Snapshot

Restore snapshot

You are creating a new DB instance or DB cluster from a snapshot. The default VPC security group and parameter group are selected for the new DB instance or DB cluster, but you can change these settings.

Free plan has access to limited features and resources

The free plan limits the features and resources that are available for RDS and Aurora databases. Upgrade your account plan to remove all limitations. [Learn more](#)

[Upgrade plan](#)

DB instance settings

DB engine

MySQL Community

☐ Enable RDS Extended Support

Amazon RDS Extended Support is a paid offering. By selecting this option, you consent to being charged for this offering if you are running your database major version past the RDS end of standard support date for that version. Check the end of standard support date for your major version in the [RDS for MySQL documentation](#).

Availability and durability

Deployment options

Choose the deployment option that provides the availability and durability needed for your use case. AWS is committed to a certain level of uptime depending on the deployment option you choose. Learn more in the [Amazon RDS service level agreement \(SLA\)](#).

Multi-AZ DB cluster deployment (3 instances)

Creates a primary DB instance with two readable standbys in separate Availability Zones. This setup provides:

- 99.99% uptime
- Redundancy across Availability Zones
- Increased read capacity
- Reduced write latency

Multi-AZ DB instance deployment (2 instances)

Creates a primary DB instance with a non-readable standby instance in a separate Availability Zone. This setup provides:

- 99.95% uptime
- Redundancy across Availability Zones

Single-AZ DB instance deployment (1 instance)

Creates a single DB instance without standby instances. This setup provides:

- 99.5% uptime
- No data redundancy

Restore Snapshot

- DB Identifier: **student-db-snapshot-restore**

Snapshot being restored

Snapshot is being restored

Restoring student-db-snapshot to student-db-snapshot-restore. This might take several minutes.

Databases (1)											
Filter by databases											
DB identifier	Status	Role	Engine	Upgrade rollout ...	Region ...	Size	Recommendations	CPU	Curren...	Mainte...	VPC
student-db-snapshot-restore	Creating	Instance	MySQL Community	SECOND	-	db.t3.micro		-		none	vpc-0e9c534171

Snapshot restored

Snapshot restored successfully

Restoring student-db-snapshot to student-db-snapshot-restore was successful.

Databases (1)											
Filter by databases											
DB identifier	Status	Role	Engine	Upgrade rollout ...	Region ...	Size	Recommendations	CPU	Curren...	Mainte...	VPC
student-db-snapshot-restore	Available	Instance	MySQL Community	SECOND	us-east-1f	db.t3.micro		8.36%	0 Conr	none	vpc-0e9c534171

Lab 8 – Monitoring Amazon RDS

CPU Utilization

- 7.22%

Summary				
DB identifier	student-db-snapshot-restore	Status	Available	Role
CPU	8.36%	Class	db.t3.micro	Instance
		Current activity	0 Connections	Engine
				MySQL Community
				Region & AZ
				us-east-1f
				Recommendations

Database Connection

- student-db-snapshot-restore.cyf4o0qgmrxxy.us-east-1.rds.amazonaws.com:3306

Connectivity & security			
Monitoring			
Logs & events			
Configuration			
Zero-ETL integrations			
Maintenance & backups			
Data migrations			
Tags			
Recommendations			
Connect using			
Code snippets			
CloudShell			
Endpoints			
Database name			
Master username			
Internet access gateway			
Port			
Endpoint type			
Additional configurations			
Connectivity & security			
Endpoint & port			
Endpoint			
Port			
Networking			
Availability Zone			
VPC			
Subnet group			
Subnets			
Network type			
Security			
VPC security groups			
Publicly accessible			
Certificate authority			
Certificate authority date			
DB instance certificate expiration date			

Free Storage Space

Primary storage

Encryption
Enabled

AWS KMS key
[aws/rds](#)

Storage type
General Purpose SSD (gp2)

Storage
20 GiB

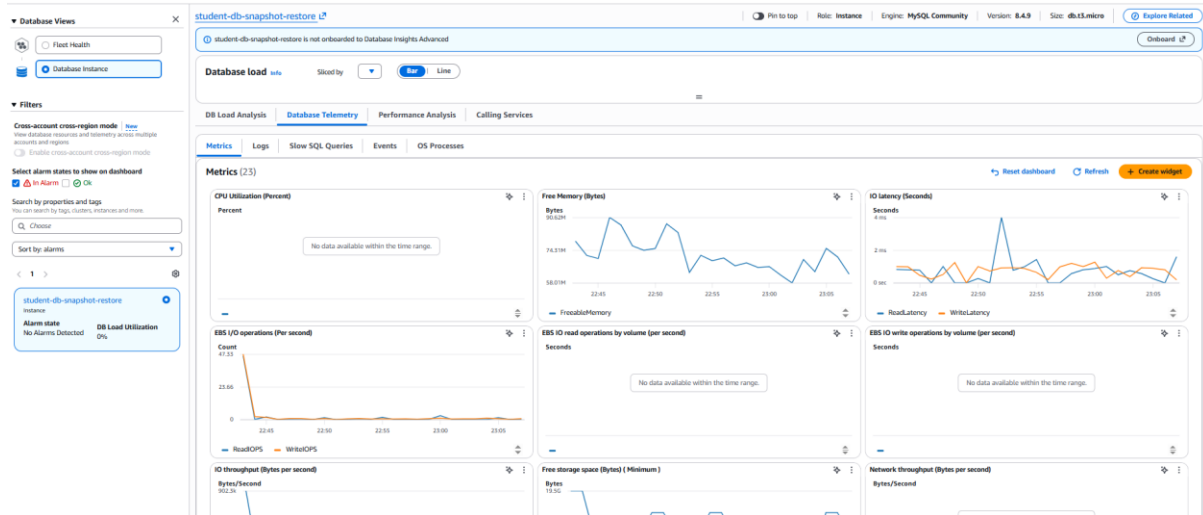
Provisioned IOPS
-

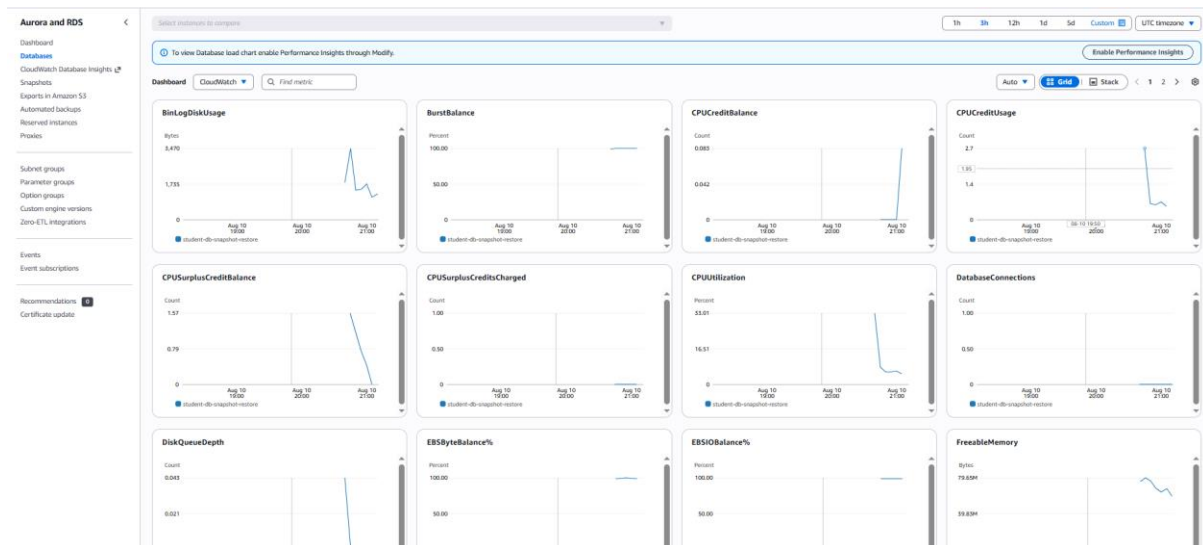
Storage throughput
-

Storage autoscaling
Disabled

Storage file system configuration
Current

Read/Write Latency





Lab 9 - Explore Amazon RDS Pricing



Lab 10